

CONSORT RANDOMIZED CLINICAL TRIAL

Evaluating Combined Effect of Oral Premedication with Ibuprofen and Dexamethasone on Success of Inferior Alveolar Nerve Block in Mandibular Molars with Symptomatic Irreversible Pulpitis: A Prospective, Double-blind, Randomized Clinical Trial



Mohit Kumar, MDS,
Rakesh Singla, MDS,
Gurdeep Singh Gill, MDS,
Tarun Kalra, MDS, and
Namita Jain, MDS

SIGNIFICANCE

A combination of preoperative ibuprofen 800 mg and dexamethasone 0.5 mg can improve the success rate of inferior alveolar nerve block in patients with symptomatic irreversible pulpitis in mandibular molars.

KEY WORDS

Dexamethasone; ibuprofen; inferior alveolar nerve block; local anesthesia; premedication; symptomatic irreversible pulpitis

Endodontic pain management mainly relies on successful local anesthesia. Inferior alveolar nerve block (IANB) is the standard injection technique to achieve local anesthesia for endodontic treatment in mandibular teeth. However, the success rate of IANB decreases drastically in patients with irreversibly inflamed pulps. Various clinical studies have reported an IANB failure rate between 43% and 83%¹⁻⁴. Pulpal inflammation often has been cited as 1 of the main causes of IANB failure in symptomatic irreversible

From the Department of Conservative Dentistry and Endodontics, JCD Dental College, Sirsa, Haryana, India

Address requests for reprints to Dr Mohit Kumar, Department of Conservative Dentistry and Endodontics, Jan Nayak Chaudhary Devi Lal Dental College, JCD Vidyapeeth, Barnala Road, Sirsa, Haryana 125055, India.

E-mail address: drmhohit2163@gmail.com
0099-2399/\$ - see front matter

Copyright © 2021 American Association of Endodontists.

<https://doi.org/10.1016/j.joen.2021.01.005>

pulpitis (SIP) of mandibular molars¹⁻⁵. In irreversible pulpitis, breakdown of damaged cell membranes and release of arachidonic acid (AA) occur. This AA is acted on by cyclooxygenase (COX) and lipoxygenase enzymes and gets converted into biologically active prostaglandins (PGs) and leukotrienes, respectively. The greater level of PGs can affect the tetrodotoxin resistance receptors and slack off the nerve response to the anesthetic solution. PGs also sensitize nerve endings to bradykinins and histamines, causing allodynia and hyperalgesia⁶⁻⁸. Leukotrienes act synergistically with PGs, thereby increasing local infiltration of neutrophils and considerably reducing the pain threshold.

One effective way of counteracting pulpal inflammation is preoperative administration of nonsteroidal anti-inflammatory drugs. Ibuprofen has been rated as the safest and most effective nonsteroidal anti-inflammatory drug in endodontic pain. Various studies have investigated the effect of preoperative ibuprofen on the success rate of IANB⁹⁻¹⁹. Although some studies have shown significant improvement⁹⁻¹³, others have reported no effect of ibuprofen on the efficacy of IANB in patients with SIP of mandibular molars¹⁴⁻¹⁹.

Glucocorticoids have been used for pain management in endodontics before, during, and after treatment because of their potent anti-inflammatory properties²⁰⁻²³. Very few endodontic studies have used preoperative dexamethasone orally or as buccal infiltration, but improved outcomes were noticed only with oral administration^{9,14,24}. Therefore, the use of preoperative dexamethasone will provide more data regarding its effect on enhancing the success of IANB.

It seems logical to assume that preoperative administration of ibuprofen and dexamethasone together will augment the anti-inflammatory effect and result in enhanced anesthetic success in teeth with irreversibly inflamed pulps. To the best of our knowledge, no study so far has evaluated their combined effect on the efficacy of IANB in mandibular molars with SIP. Therefore, the aim of this study was to determine the combined effect of preoperative ibuprofen and dexamethasone on the success rate of IANB in cases of mandibular molars with SIP. The null hypothesis was that there will be no effect of the combination of ibuprofen and dexamethasone on the success of IANB in mandibular molars with SIP.

MATERIALS AND METHODS

This prospective, double-blind, randomized clinical study was approved by the Institutional

Review and Ethical Committee of JCD Dental College, Sirsa, India (approval number JCDV/DC/18/2152). Ninety-four adult patients were randomly selected from among the emergency patients referred to the Department of Conservative Dentistry and Endodontics. Written informed consent was obtained from each patient. All included patients were experiencing moderate to severe pain (55–170 mm readings on the Heft-Parker visual analog scale [VAS]) in mandibular molars, had a normal periapical radiographic appearance, were in good health (American Society of Anesthesiologists class I or II), and had not taken any analgesics for at least 12 hours before enrollment in the study. Patients under 18 years old; patients with allergies and hypersensitivity to medications, local anesthetic solution, or its components used in the study; pregnant or nursing women; immunocompromised patients; patients with active sites of pathosis in the area of injection; and patients unable to give consent were excluded from the study. The clinical diagnosis of SIP was confirmed with electric pulp testing (Digitest; Parkell Inc, Brentwood, NJ) and cold testing (Endo Ice 1,1,1,2 tetrafluoroethane; Coltene, which resulted in an early and prolonged painful response).

For pain evaluation, the Heft-Parker VAS was used in this study²⁵. This is a 170-mm scale divided into 4 categories: no pain (0 mm), mild pain (0–54 mm), moderate pain (55–114 mm), and severe pain (115–170 mm). Patients were properly educated for markings on the VAS. All patients recruited in the study had given 4 readings for pain during the entire procedure (ie, before premedication, after local anesthesia, during access preparation, and during file placement in the canals).

On the basis of the pilot study with a confidence interval of 95% and an alpha error of 5%, the minimum sample size per group was calculated, which turned out to be 20. Considering a 10% dropout rate, the minimum sample size for the study was kept at 90.

Both the patient and operator were blinded in this study. A dedicated staff member was given the responsibility of preparing, coding, and administering premedications to the patient. The premedications (ie, placebo [lactose powder], dexamethasone, and ibuprofen) were dispensed in identically appearing capsules and assigned a 4-digit unique code. Using a roll of dice, patients were randomly allocated to 4 different groups and received the following: placebo (placebo group [PLAg]), 0.5 mg dexamethasone (dexamethasone group [DEXg]), 800 mg ibuprofen (ibuprofen group [IBUg]), and a combination of 0.5 mg dexamethasone and 800 mg ibuprofen (combination group

[COMg]). The operator was unaware of the prescribed medication or the group to which the patient was allocated. The operator only received the 4-digit assigned code, which was transferred to the history sheet of the patient. The unique code was broken only after the completion of the study. Except for administering premedications, all the procedures were performed by a single operator. After 1 hour of oral premedication, all patients received standard IANB with 1.8 mL 2% lignocaine containing 1:200,000 adrenaline (Cadila Pharmaceuticals Limited, New Delhi, India) via a 3-mL disposable syringe with a 25-G needle (Dispovan; HMD, Faridabad, India). Thereafter, every patient was questioned for lip numbness. If deep lip numbness was not reported by 15 minutes of local anesthesia, the patient was excluded from the study. Patients who achieved lip numbness recorded their pain scores, and access preparation was initiated under rubber dam isolation. No additional form of anesthesia was used to numb the buccal gingiva for clamp placement. None of the patients reported any discomfort during clamp placement. Access preparation was performed using the Endo Access Kit (Dentsply Sirona, York, PA); canal orifices were located with the Endo Explorer DG 16 (GDC, Hoshiarpur, India); and stainless steel K-files #06, 08, and 10 (Mani, Tochigi, Japan) were used for the initial instrumentation of the root canals.

Patients were instructed to raise their hand if they felt pain during the endodontic procedure. If the pain was moderate to severe (during dentin penetration or intracanal file placement), the procedure was immediately stopped, and patients rated their pain on the Heft-Parker VAS. Success was defined as no or mild pain (0–54 mm), whereas moderate to severe pain (55–170 mm) anytime during the procedure was considered as failure. All failure cases were managed by the same operator using supplemental local anesthesia techniques, and treatment was completed in the same or subsequent appointments.

REFERENCES

1. Cohen HP, Cha BY, Spangberg LS. Endodontic anesthesia in mandibular molars: a clinical study. *J Endod* 1993;19:370–3.
2. Kennedy S, Reader A, Nusstein J, et al. The significance of needle deflection in success of the inferior alveolar nerve block in patients with irreversible pulpitis. *J Endod* 2003;29:630–3.
3. Claffey E, Reader A, Nusstein J, et al. Anesthetic efficacy of articaine for inferior alveolar nerve blocks in patients with irreversible pulpitis. *J Endod* 2004;30:568–71.
4. Bigby J, Reader A, Nusstein J, et al. Anesthetic efficacy of lidocaine/meperidine for inferior alveolar nerve blocks in patients with irreversible pulpitis. *J Endod* 2007;33:7–10.
5. Hargreaves KM, Keiser K. Local anesthetic failure in endodontics: mechanisms and management. *Endod Topics* 2002;1:26–39.
6. Shirvani A, Shamszadeh S, Eghbal MJ, et al. Effect of preoperative oral analgesics on pulpal anesthesia in patients with irreversible pulpitis-a systematic review and meta-analysis. *Clin Oral Investig* 2017;21:43–52.
7. Renton T, Yiangou Y, Baecker PA, et al. Capsaicin receptor VR1 and ATP purinoceptor P2X3 in painful and non-painful human tooth pulp. *J Orofac Pain* 2002;17:245–50.
8. White RK, Montgomery S. Leukotrienes: inflammatory mediators - a review. *Oral Surg Oral Med Oral Pathol Oral Radiol* 1986;61:514–8.
9. Bidar M, Mortazavi S, Forghani M, et al. Comparison of effect of oral premedication with ibuprofen or dexamethasone on anesthetic efficacy of inferior alveolar nerve block in patients with irreversible pulpitis: a prospective, randomized, controlled, double-blind study. *Bull Tokyo Dent Coll* 2017;58:231–6.
10. Noguera-Gonzalez D, Cerda-Cristerna BI, Chavarria-Bolaños D, et al. Efficacy of preoperative ibuprofen on the success of inferior alveolar nerve block in patients with symptomatic irreversible pulpitis: a randomized clinical trial. *Int Endod J* 2013;46:1056–62.
11. Modaresi J, Dianat O, Mozayeni MA. The efficacy comparison of ibuprofen, acetaminophen-codeine, and placebo premedication therapy on the depth of anesthesia during treatment of inflamed teeth. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 2006;102:399–403.
12. Parirokh M, Ashouri R, Rekabi AR, et al. The effect of premedication with ibuprofen and indomethacin on the success of inferior alveolar nerve block for teeth with irreversible pulpitis. *J Endod* 2010;36:1450–4.
13. Shantiaee Y, Javaheri S, Movahhedian A, et al. Efficacy of preoperative ibuprofen and meloxicam on the success rate of inferior alveolar nerve block for teeth with irreversible pulpitis. *Int Dent J* 2017;67:85–90.

14. Shahi S, Mokhtari H, Rahimi S. Effect of premedication with ibuprofen and dexamethasone on success rate of inferior alveolar nerve block for teeth with asymptomatic irreversible pulpitis: a randomized clinical trial. *J Endod* 2013;39:160–2.
15. Oleson M, Drum M, Reader A, et al. Effect of preoperative ibuprofen on the success of the inferior alveolar nerve block in patients with irreversible pulpitis. *J Endod* 2010;36:379–82.
16. Ianiro SR, Jeanson BG, McNeal SF, et al. The effect of preoperative acetaminophen or a combination of acetaminophen and ibuprofen on the success of inferior alveolar nerve block for teeth with irreversible pulpitis. *J Endod* 2007;33:11–4.
17. Simpson M, Drum M, Nusstein J, et al. Effect of combination of preoperative ibuprofen/acetaminophen on the success of the inferior alveolar nerve block in patients with symptomatic irreversible pulpitis. *J Endod* 2011;37:593–7.
18. Aggarwal V, Singla M, Kabi D. Comparative evaluation of effect of preoperative oral medication of ibuprofen and ketorolac on anaesthetic efficacy of inferior alveolar nerve block with lidocaine in patients with irreversible pulpitis: a prospective, double-blind, randomized clinical trial. *J Endod* 2010;36:375–8.
19. Jena A, Shashirekha G. Effect of preoperative medications on the efficacy of inferior alveolar nerve block in patients with irreversible pulpitis: a placebo-controlled clinical study. *J Conserv Dent* 2013;16:171–4.
20. Pochapski MT, Santos FA, Dias de Andrade E, Sydney GB. Effect of pretreatment dexamethasone on postendodontic pain. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 2009;108:790–5.
21. Liesinger A, Marshall FJ, Marshall JG. Effect of variable doses of dexamethasone on post-treatment endodontic pain. *J Endod* 1993;19:35–9.
22. Mehrvarzfar P, Esnashari E, Salmanzadeh R. Effect of dexamethasone intraligamentary injection on post-endodontic pain in patients with symptomatic irreversible pulpitis: a randomized controlled clinical trial. *Iran Endod J* 2016;11:261–6.
23. Krasner P, Jackson E. Management of posttreatment endodontic pain with oral dexamethasone: a double-blind study. *Oral Surg Oral Med Oral Pathol* 1986;62:187–90.
24. Aggarwal V, Jain A, Kabi D. Anesthetic efficacy of supplemental buccal and lingual infiltrations of articaine and lidocaine after an inferior alveolar nerve block in patients with irreversible pulpitis. *J Endod* 2009;35:925–9.
25. Heft MW, Parker SR. An experimental basis for revising the graphic rating scale for pain. *Pain* 1984;19:153–61.
26. Marshall JG. Consideration of steroids for endodontic pain. *Endod Topics* 2002;3:41–51.
27. Tripathi KD. Nonsteroidal anti-inflammatory drugs and antipyretic-analgesics. In: Tripathi KD, editor. *Essentials of Medical Pharmacology*. 7th ed. New Delhi, India: Jaypee; 2013. p. 192–209.
28. Huang ZJ, Hsu E, Li HC, et al. Topical application of compound ibuprofen suppresses pain by inhibiting sensory neuron hyperexcitability and neuroinflammation in a rat model of inter-vertebral foramen inflammation. *J Pain Res* 2011;12:141–52.
29. Cummins TR, Waxman SG. Sodium channels in pain pharmacology. In: Beaulieu P, Lussier D, Porreca F, Dickenson AH, editors. *Pharmacology of Pain*. Seattle, WA: IASP Press; 2010. p. 139–62.
30. Mahajan P, Singh G, Kaur R, et al. A comparative clinical study to evaluate the effect of premedication with ibuprofen, tramadol and combination of ibuprofen and acetaminophen on success of inferior alveolar nerve block in patients with asymptomatic irreversible pulpitis. *Bangladesh J Med Sci* 2017;16:370–4.
31. Eldin S, Roshdy NN, Fouda MY. A comparative assessment of the effect of pretreatment dexamethasone versus placebo on post-endodontic pain and success of inferior alveolar nerve block in mandibular molars with symptomatic irreversible pulpitis: a blinded randomized clinical trial therapeutic study. *Acta Sci Dent Sci* 2020;4:60–6.
32. Briggs M, Closs JS. A descriptive study of the use of visual analogue scales and verbal rating scales for the assessment of post-operative pain in orthopedic patients. *J Pain Symptom Manage* 1999;18:438–46.
33. Ong CK, Lirk P, Tan CH, et al. An evidence-based update on nonsteroidal anti-inflammatory drugs. *Clin Med Res* 2007;5:19–34.
34. Zanjir M, Lighvan NL, Yarascavitch C, et al. Efficacy and safety of pulpal anesthesia strategies during endodontic treatment of permanent mandibular molars with symptomatic irreversible pulpitis: a systematic review and network meta-analysis. *J Endod* 2019;45:1435–64.